

123 years of MLB: How Has Winning Percentage Changed Since 1901?



Raina Duffy | CSCI 120 ---- Spring 2026

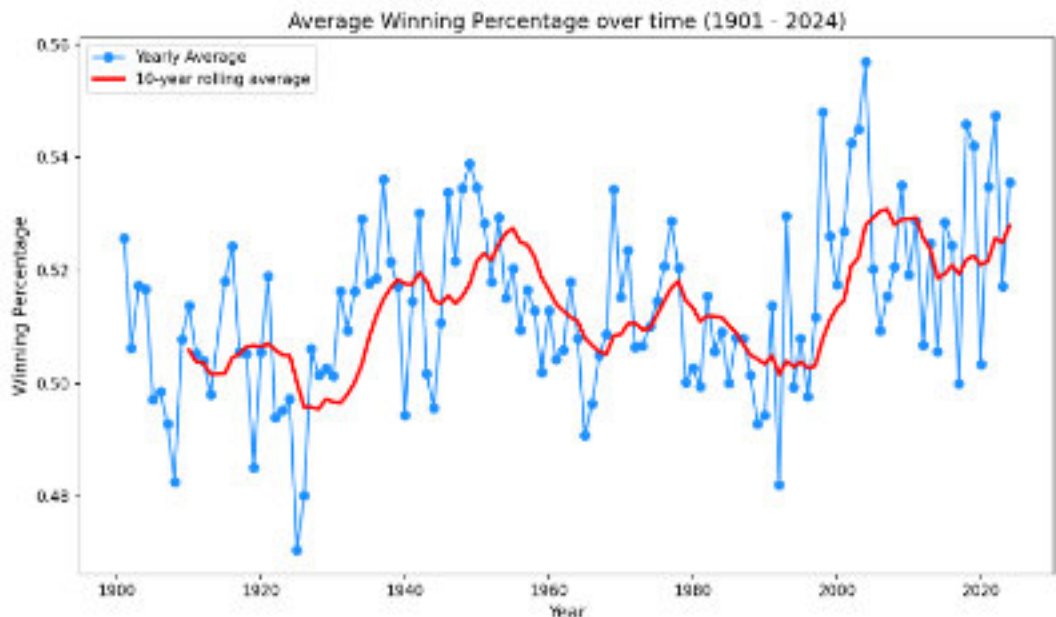
1 Introduction

Baseball has evolved dramatically since 1901; from the dead-ball era to the steroid era to modern analytics. This project investigates whether those changes translated into measurable improvement in team performance, or whether winning percentage has remained essentially stable across 123 years of Major League Baseball.

2 Data & Methods

- Source: Kaggle
- InitialDataset: 2690 rows, 16 columns
- Cleaned: 1168 rows, 9 columns
- Key Variables: team_name, year, wins, winning_percentage, run_differential, runs_scored, runs_allowed
- Approach: Correlation

3 Chart 1 --Trend Over Time



Average winning percentage per year from 1901 to 2024. The red line shows the 10-year rolling average, revealing a stable trend with a slight uptick post-2000.

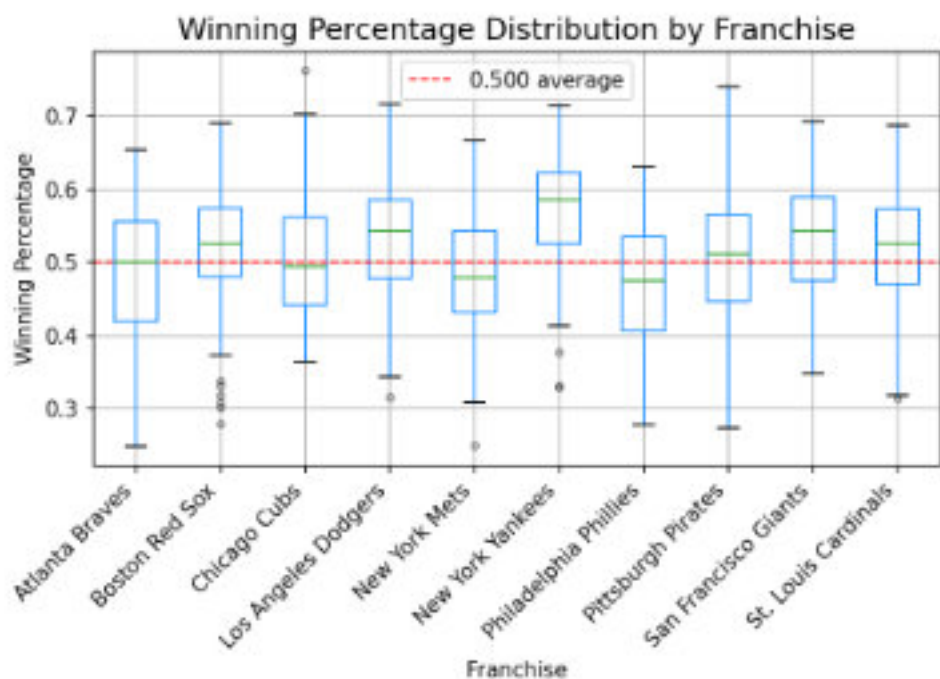
4 Key Findings

- The Yankees averaged 0.570, while the Phillies averaged 0.467, a gap of 0.10 across 123 seasons.
- Overall average Winning Percentage went from 0.526 in 1901 to 0.536 in 2024, a change of only 0.01 over 123 years.
- Correlation between Year and Winning Percentage are weakly correlated ($r=0.065$, $p=0.027$)
- Though a pattern of ($r=0.947$, $p=0.0$) confirms a strong correlation between Winning Percentage and Run Differential

5 Charts 2 + 3



Average winning percentage by franchise. The Yankees lead at 0.570 while the Phillies trail at 0.467. Red line marks 0.500.



All teams show wide year-to-year variation with medians near 0.500.

6 Conclusion

The average winning percentage of MLB teams has not meaningfully changed over 123 years ($r=0.065$, $p=0.027$). Despite major shifts in the sport including professionalization, expansion, the steroid era, and modern analytics. Winning Percentage has remained close to 0.500 across all franchises from 1901 to 2024. My hypothesis was not supported. Improvement in one franchise comes at another's expense in baseball.

Data Source

"MLB Statistics" Kaggle 2025. <https://www.kaggle.com/datasets/diazk2/mlb-statistics-1901-present>